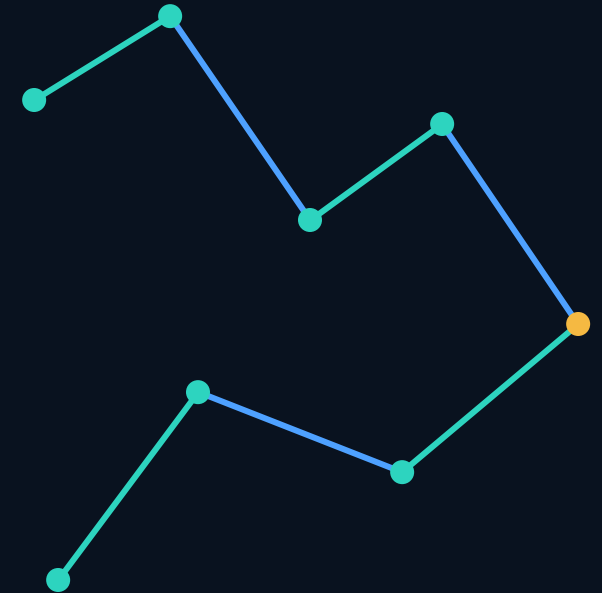


EWMA Covariate Shift Demo

Research progress review

Auditable CSE reconstruction for motor-imagery EEG
Classifier benchmarks and complementary PCA monitoring

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5th October 2026

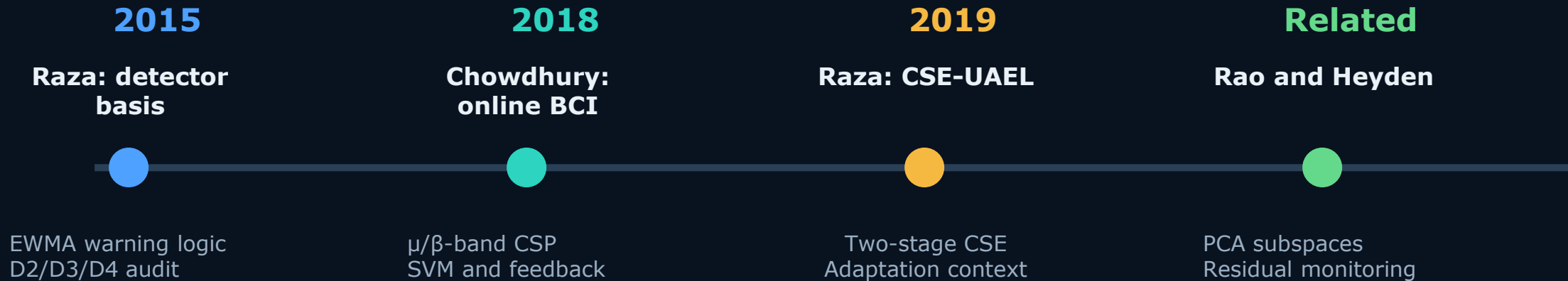


Detection and adaptation are implemented

Workstream	Current position
CSE reconstruction	EEG features, EWMA warnings and Stage-II diagnostics
Supervised adaptation	Predict first, then update using the revealed label
Transductive adaptation	PWKNN pseudo-labels; confidence-gated training updates
Policy experiments	Five Diethe-inspired policies on matched streams
Evidence still needed	Matched reproduction runs and saved adaptation comparisons

Full dynamic CSE-UAEL ensemble growth and weighted voting remain open.

Reading informs the reconstruction and its limits



Diethe et al. (2019)

Monitoring, update policies and adaptation cost

Heyden et al. (2024): reconstruction loss basis

$$L_t = \text{MSE}(X_t, \hat{X}_t) = \frac{1}{d} \sum_{j=1}^d (X_{t,j} - \hat{X}_{t,j})^2$$

Alippi et al. (2017): hierarchical detection provides a related framework for warning and validation.

Chowdhury et al. (2018): online adaptive BCI

Calibration and task

10 healthy participants + 10 stroke patients
80 calibration trials, then 40 online trials
Left- versus right-hand motor imagery
Initial linear SVM and CSD parameters

Signal and adaptation

μ 8–12 Hz and β 16–24 Hz
CSP yields four log-variance features
PC1 EWMA warning, then Hotelling T^2
Validated shift triggers SVM retraining

Online output

Visual feedback and hand-exoskeleton movement follow MI classification.

Comparator: EEG-NAC replays the same recorded data offline with a fixed classifier.

Source: §II.B, p. 1072 · §II.D–E, pp. 1073–1075 · Figs. 1, 3 and 4

REPORTED ACHIEVEMENTS

Decoding improved relative to the fixed classifier

Group	EEG-NAC	EEG-CSAC	Gain
Healthy (n = 10)	75.25 ± 5.46%	81.50 ± 4.89%	+6.25 pp
Stroke (n = 10)	70.25 ± 3.43%	75.75 ± 3.92%	+5.50 pp

Both accuracy comparisons: $p < 0.01$

Kappa also improved significantly. Every participant could trigger the exoskeleton.

Source: §III, p. 1076, Tables II–IV and Fig. 6 · §IV, p. 1078

DETECTOR EVIDENCE

Stage-II validation reduced retraining triggers

Group	Stage-I CSW	Stage-II CSV
Healthy	15.75 ± 5.65%	9.25 ± 3.54%
Stroke	16.75 ± 2.89%	9.00 ± 2.10%

Rates are percentages of trials. Both reductions: $p < 0.01$.

Validated-shift rate and accuracy gain correlate: $r = 0.853$, $p < 0.01$.

Interpretation: fewer retraining triggers do not independently quantify false alarms.

Source: §II.E.4, p. 1075, Eqs. (10)–(11) · §III, p. 1077, Tables VI–VIII and Fig. 7

Limitations acknowledged by the authors

Representation and comparison

Binary left/right MI classification
Fixed μ/β bands for all participants
No direct test against other adaptive BCIs

Labels and online control

Supervised adaptation assumes cue compliance
Synchronous BCI, one feedback event per trial
Continuous/asynchronous control remains future work

Artefacts and external noise

Online preprocessing adds latency. The authors acknowledge that contaminated EEG can reduce decoding accuracy.

Source: §IV, p. 1078 (scope, frequency bands, comparator, noise) · pp. 1078–1079 (control and labels)

Limits of the experimental evidence

Small cohort and short follow-up

10 participants in each group
One 40-trial online feedback run
Limited evidence across patient subgroups
and repeated rehabilitation sessions

Comparison and recovery

Fixed baseline applied offline to the
same data
No parallel online nonadaptive condition
Reported metrics concern BCI
performance
Durable motor recovery remains
untested

Supported conclusion

The study establishes online technical feasibility and improved decoding.
Longer-term reliability and rehabilitation benefit require further evaluation.

Source: §II.B, p. 1072 · §II.D–E, p. 1073 · §III, pp. 1075–1077

Relevance to the current EWMA project

Methodological connection

EWMA warning followed by shift validation
Selective adaptation motivates event auditing
Paper uses two bands and four CSP features
Current deck uses 10-band FBCSP on 2A/2B

Existing next-step priorities

Lock the reproduction configuration
Explain accepted and rejected Stage-II events
Measure synthetic false alarms and delay
Export real EEG score/residual traces

Evidence boundary

Keep the paper's online results separate from repository benchmarks. Numerical reproduction and real EEG extension evidence remain open in the current deck.

Sources: Chowdhury et al. (2018), §II.E and §III–IV; repository review at f3ef2e8.

How evidence becomes an update decision

Responsibility	Implemented behaviour
Representation	Cue-aligned EEG; FBCSP; PCA fitted on training data
Detection and validation	EWMA warnings; explicit Stage-II interpretations
Update policy	Never, periodic, warning, validated or performance-drop
Label regime	Supervised labels or confidence-gated PWKNN pseudo-labels
Audit outputs	Predictions, accepted/rejected evidence and update events

Score-plus-residual monitoring is a separate Stage-I extension.

The Diethe comparison currently uses the fixed CSE detector.

Repository evidence: 2a8d20b. Implemented capabilities and stored results are distinguished.

From EEG signals to the monitored PCA score

Two frequency bands: μ 8–12 Hz and β 16–24 Hz. Fit the transformations on calibration data.

01 Spatial filtering

$$Z = WE$$

Combine EEG channels using learned CSP filters.

E: one EEG trial; W: spatial filters; Z: filtered signals.

02 Log-variance features

$$f_j = \log \left(\frac{\text{Var}(Z_j)}{\sum_{k=1}^{2m} \text{Var}(Z_k)} \right)$$

Express each selected signal's variance relative to the total.

The paper retains two features per band: four per trial.

03 PCA and PC1 selection

$$\mathbf{s}_i = P^T(\mathbf{f}_i - \boldsymbol{\mu}_{\text{cal}}), \quad x_i = s_{i,1}$$

Transform the feature vector, then monitor its first PC score.

The classifier uses features; Stage I monitors the scalar x_i .

Chowdhury et al. (2018), §II.E.1 and §II.E.3, Eqs. (1), (2), (4), pp. 1073–1074.

Indices clarified; PCA shown with explicit calibration centring. Code: fbcsp.py; detection/preprocessing.py.

How PCA scores become warnings and confirmations

Stage I: EWMA warning

State and one-step prediction error

$$z_i = \lambda x_i + (1 - \lambda)z_{i-1}, \quad e_i = x_i - z_{i-1}$$

Control limits use the previous variance

$$UCL_i = z_{i-1} + L\sigma_{e,i-1}, \quad LCL_i = z_{i-1} - L\sigma_{e,i-1}$$

Smoothed error variance • Eq. (8)

$$\sigma_{e,i}^2 = \alpha_v e_i^2 + (1 - \alpha_v)\sigma_{e,i-1}^2$$

Check current limits first; update state and variance afterwards.

α_v is the variance-smoothing weight (paper: 0.05).

Chowdhury §II.E.3, Eqs. (3), (5)–(9). Eq. (9) uses separate covariance terms; pooled mode is a repository choice. Eq. (8) describes the always-update variance mode. The Diethe pilot uses frozen variance.

Stage II: Hotelling validation

Repository pooled two-sample mode

$$T^2 = \frac{n_1 n_2}{n_1 + n_2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^T S_p^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$$

Decision and dimensionality condition

$$p < \alpha_{\text{test}}, \quad 2H - d - 1 > 0$$

H: observations per window; d: retained dimensions.

α_{test} is distinct from variance smoothing.

Confirmation is available after the second window.

CSE development and held-out evaluation

BCI Competition IV 2A

Session I develops the model

Stratified 70/30 split within Session I
Fit FBCSP and PCA on training only
Transform validation with fitted models
Hold out Session II, left/right trials

BCI Competition IV 2B

Sessions I–III develop the model

Stratified 70/30 split within Sessions I–III
Fit FBCSP and PCA on training only
Transform validation with fitted models
Hold out Sessions IV–V

Shared rule

**Evaluation trials do not enter feature-model fitting.
Record the split seed and effective detector configuration.**

Classifier benchmark: 2A fits Session I and tests II.
The stored 2B benchmark fits I–II and tests labelled Session III.

Classifier and complementary-monitoring results

Real EEG classifier benchmark

Mean accuracy across 9 subjects per dataset

Dataset	Single SVM	Bagged SVM
2A	76.00%	75.85%
2B	74.86%	75.42%

Mean change: -0.15 pp (2A), +0.56 pp (2B).
No consistent gain across both datasets.

2A: Session II evaluation
2B: Session III GDF-only holdout

CI at 2a8d20b: 255 passed, 1 failed. Homepage test expects /diethe; route is /results/diethe.

Complementary Stage-I monitoring

Detection within 20 samples, 100 seeds, magnitude 3

Shift	Score	Residual	Combined
PC1	35%	21%	31%
PC2	21%	100%	100%
Both	26%	100%	99%

Magnitude 6: PC1 detection rises to 88%
(score) and 82% (combined).

No-shift 20-sample alarm rate: 21-26%.
Per-sample false-alarm rate: about 1%.

$Q_t = dL_t$

B01: two saved runs give different validation totals

B01 evidence	Warnings	Validated shifts
Published reference	14	10
Standalone diagnostic	17	5
Combined Table-I file	17	0

What this means

Both stored runs differ from the published result. Neither file proves that one run supersedes the other.

Next: record the Stage-II mode, window settings and generating commit for each run.

Sources: `outputs/metrics/bci_2b_table1_results.csv` and `bci_table1_comparison.csv` at 2a8d20b.

Numerical reproduction remains unresolved

Mean validated-shift count	Computed	Published
Dataset 2A	0.33	5.56
Dataset 2B	0.67	10.33

Evidence available

Subject-level comparisons

Classifier benchmarks

Synthetic score/residual ablations

Evidence to complete

One configuration per reproducible run

Stage-II acceptance/rejection analysis

Real EEG traces and repeated Diethe runs

Source: outputs/metrics/bci_table1_comparison.csv at 2a8d20b; means calculated across nine subjects per dataset.

Dataset roles differ across experiments

Experiment	2A	2B
Table-I CSE	Session I development; Session II evaluation	I–III development; IV–V evaluation
Stored classifier benchmark	Fit I; evaluate II	Fit I–II; evaluate III
Complementary EEG runner	I: training/calibration split; II evaluation	Fit I–II; calibrate III; evaluate IV–V
Diethe policy laboratory	Synthetic feature streams by default; prepared EEG features accepted by CLI	Raw EEG loading is not automatically integrated

Real EEG alarm traces describe alert burden. They do not supply ground-truth shift times.

Source: current runners and stored result files at 2a8d20b. Evaluation data do not fit the feature representation.

Diethe: separate change, performance and action

Question	Repository experiment
Has the input changed?	Shared CSE warnings and Stage-II validation records
Has prediction quality changed?	Rolling accuracy against a held-out validation reference
Should the model update?	Five explicit rules compared on the same stream
What does updating cost?	Update count, fitting time, prediction time and training size

First five-policy run is saved. Repeated comparisons across scenarios and seeds remain pending.

The five-policy comparison is a repository experiment inspired by the paper.

Sources: Diethe et al. (2019), §§3–5; docs/diethe_experiments.md; src/adaptation/experiments/diethe.py.

Five policies, three controlled scenarios

Policy	Update trigger
Never / periodic	No update / every configured interval
Warning / validated	Stage-I warning / available Stage-II confirmation
Performance drop	Rolling accuracy drops below the validation reference

No shift: baseline alert and update burden.

Feature shift: $P(X)$ changes; the label rule stays fixed.

Relationship shift: the label rule changes; $P(X)$ stays fixed.

Predict before revealing each label. Training expands after updates.

Delayed labels, rollback and energy measurement remain outside this experiment.

Source: docs/diethe_experiments.md and current CLI. Defaults: 240 trials, 20-trial interval, 30-trial performance window.

A detected shift does not guarantee a useful update

Synthetic feature shift; seed 42; 240 trials; shift starts at index 120.

Policy	Accuracy	Updates	Gain vs never
Never	97.92%	0	0.00 pp
Periodic	98.33%	12	+0.42 pp
Warning	97.50%	3	−0.42 pp
Validated	97.50%	1	−0.42 pp
Performance drop	97.92%	0	0.00 pp

Periodic updating gains one correct prediction overall for 12 updates.

Validated updating finishes one prediction below never-update.

One-run illustration, not a policy ranking. Next: matched comparisons across scenarios and seeds.

Source: `outputs/diethe/20261004T111759599003Z/summary.csv`; generating commit 2a8d20b.

First-year objectives across twelve months

Proposed October 2026 to September 2027 plan. Existing outputs provide the initial baseline.

WORK PACKAGE	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	EXPECTED OUTPUT
WP1 Theory and architecture CPD, CL and twin design													Architecture specification
WP2 CPD benchmarking EWMA, CUSUM, BOCPD, density ratios													Benchmark suite and report
WP3 Continual learning and sustainability Replay, forgetting, energy costs													CL module and cost analysis
WP4 Digital twin co-evolution Disagreement, joint updates, safety													Twin framework and testbed
WP5 Cross-domain evaluation BCI and photonic maintenance													Compare retraining and periodic updates
WP6 Synthesis and engagement Manuscripts, release, stakeholders													Submissions and adoption roadmap

October priority: lock reproduction settings, explain Stage-II gaps and export real EEG complementary traces.

Next milestone: a reproducible evidence package

1. Reconcile the saved reproduction runs

Lock settings and explain the Stage-II accepted, rejected and ineligible events.

2. Generate adaptation and real EEG evidence

Extend the saved Diethe pilot across three scenarios and 20 seeds. Export 2A/2B complementary traces.

3. Align tests, slides and research notes

Resolve the homepage route assertion. Refresh Notion implementation status and link every numerical claim to its run.